PHYS 193 · Chapter 30 · Faraday's Law — self-check questions

Ten questions, two per section (§30.1–30.5), average level. Answer on paper, then check the key on the last page. Serway & Jewett 11e · Dr. K. Khelifa-Kerfa · Qatar University · Fall 2026

§30.1 · Faraday's law of induction

1. A circular loop of radius 10.0 cm lies with its plane perpendicular to a uniform magnetic field. The field increases steadily from 0.20 T to 0.60 T in 0.50 s. What is the magnitude of the average emf induced in the loop?

(a) 6.3 mV
(b) 25 mV
(c) 50 mV
(d) 0.13 V

§30.1 · Faraday's law of induction

2. A flat coil sits in a uniform, steady magnetic field. It is rotated from $\theta = 0°$ (normal along B) to $\theta = 90°$ in a time Δt, and then held fixed at 90°. Which statement is correct?

(a) An emf is induced only while the coil is rotating.
(b) The emf is largest once the coil is held at 90°, because the flux is then zero.
(c) No emf is induced at any time, because B is constant.
(d) The emf is constant for as long as $\theta \neq 0$.

§30.2 · Motional emf

3. A metal rod 0.50 m long moves at 4.0 m/s perpendicular to a uniform 0.25 T field, with its length perpendicular to both the velocity and the field. The potential difference between its ends is

(a) 0.031 V
(b) 0.13 V
(c) 0.50 V
(d) 2.0 V

§30.2 · Motional emf

4. A bar of length 0.30 m slides on frictionless rails through a 0.40 T field (into the page) at a constant 5.0 m/s. The circuit resistance is 2.0 Ω. What power is delivered to the resistor?

(a) 0.090 W
(b) 0.18 W
(c) 0.36 W
(d) 0.60 W

§30.3 · Lenz's law

5. The north pole of a bar magnet is pushed toward a circular loop of wire. Viewed from the magnet's side, the induced current in the loop circulates

(a) clockwise
(b) counterclockwise
(c) first one way, then the other
(d) not at all — the loop is not connected to a battery

§30.3 · Lenz's law

6. A loop lies in a magnetic field directed into the page. The field is being switched off, so its magnitude decreases. The current induced in the loop is

(a) clockwise, producing a field into the page inside the loop
(b) counterclockwise, producing a field out of the page inside the loop
(c) zero, because the field is still into the page
(d) clockwise, producing a field out of the page

§30.4 · The general form of Faraday's law

7. A long solenoid of radius R = 3.0 cm carries a current that makes its interior field change at dB/dt = 50 T/s. What is the magnitude of the induced electric field at r = 2.0 cm from the axis?

(a) 0.25 V/m
(b) 0.50 V/m
(c) 0.75 V/m
(d) 1.0 V/m

§30.4 · The general form of Faraday's law

8. Which statement about the electric field induced by a changing magnetic field is correct?

(a) It exists only inside a conducting loop.
(b) Its line integral around any closed path is zero, like the field of static charges.
(c) It is non-conservative and can exist at points where B itself is zero.
(d) It can be described by an electric potential $V = -\int E \cdot ds$.

§30.5 · Generators and motors

9. A generator coil has 100 turns of area 0.050 m^2 and rotates at 60 rev/s in a uniform 0.20 T field. Its maximum emf is

(a) 1.0 V
(b) 60 V
(c) 3.8×10^2 V
(d) 6.0×10^2 V

§30.5 · Generators and motors

10. A DC motor is connected to a 120 V supply. Its coil has a resistance of 10 Ω, and at its normal running speed the back emf is 70 V. The current it draws at normal speed is

(a) 1.2 A
(b) 5.0 A
(c) 7.0 A
(d) 12 A

Content generated by Claude (Anthropic) and verified by Dr. K. Khelifa-Kerfa

Answer key and reasoning

Quick key: 1. (b) 2. (a) 3. (c) 4. (b) 5. (b) 6. (a) 7. (b) 8. (c) 9. (c) 10. (b)

1. (b) 25 mV — $\Delta\Phi_B = \Delta B \cdot A = (0.40\ \text{T})(\pi \times 0.100^2\ \text{m}^2) = 1.26 \times 10^{-2}$ Wb, so $|\mathcal{E}| = \Delta\Phi_B/\Delta t = 1.26 \times 10^{-2} / 0.50 = 2.5 \times 10^{-2}$ V = 25 mV. Only the *change* of B matters, not its value.

2. (a) An emf is induced only while the coil is rotating. — $\Phi_B = BA\cos\theta$ changes while θ changes; once the coil is held, Φ_B is constant (zero) and $d\Phi_B/dt = 0$. A constant B does not mean a constant flux — θ is one of the three knobs.

3. (c) 0.50 V — Motional emf: $\Delta V = B\ell v = (0.25\ \text{T})(0.50\ \text{m})(4.0\ \text{m/s}) = 0.50$ V (Eq. 30.4). The magnetic force on the free electrons piles them up at one end until qE = qvB.

4. (b) 0.18 W — $\mathcal{E} = B\ell v = 0.40 \times 0.30 \times 5.0 = 0.60$ V; $I = \mathcal{E}/R = 0.30$ A; $P = I^2R = (0.30)^2(2.0) = 0.18$ W. The same 0.18 W is $F_{app}v = (I\ell B)v = (0.036\ \text{N})(5.0\ \text{m/s})$: the agent pushing the bar supplies it (Eq. 30.7).

5. (b) counterclockwise — The flux from the approaching N pole increases, so the induced current must create a field pointing back toward the magnet (a north face toward the magnet, repelling it). Seen from the magnet, a field pointing toward the viewer comes from a counterclockwise current (right-hand rule).

6. (a) clockwise, producing a field into the page inside the loop — Lenz: the induced current opposes the *change*. The flux into the page is decreasing, so the induced field inside the loop points into the page — along the external field, to keep the flux up. A clockwise current produces a field into the page.

7. (b) 0.50 V/m — Inside the solenoid (r < R), Faraday's law in the form $\oint E \cdot ds = -d\Phi_B/dt$ with circular symmetry gives $E(2\pi r) = \pi r^2\, dB/dt$, so $E = (r/2)\, dB/dt = (0.010\ \text{m})(50\ \text{T/s}) = 0.50$ V/m (Example 30.6b).

8. (c) It is non-conservative and can exist at points where B itself is zero. — $\oint E \cdot ds = -d\Phi_B/dt \neq 0$: the induced field is non-conservative and has no potential function. Its field lines close on themselves, and it exists wherever the flux through the path changes — including outside a solenoid, where B = 0 (Pitfall Prevention 30.1).

9. (c) 3.8 × 10² V — $\omega = 2\pi(60\ \text{rev/s}) = 377$ rad/s. $\mathcal{E}_{max} = NBA\omega = (100)(0.20\ \text{T})(0.050\ \text{m}^2)(377\ \text{rad/s}) = 377\ \text{V} \approx 3.8 \times 10^2$ V (Eq. 30.11). The emf varies as $\mathcal{E}_{max} \sin \omega t$.

10. (b) 5.0 A — $I = (\Delta V - \mathcal{E}_{back})/R = (120 - 70)/10 = 5.0$ A. If the motor stalls, the back emf vanishes and the current rises to 120/10 = 12 A — the heating I^2R goes up by a factor of $(12/5)^2 \approx 5.8$, which is why a jammed motor burns out (Example 30.8).